Knowledge Test of 23/07/2020 [1]

Exercise 1

Let $a > 0$ and X a random variable related to the following function $f_X(x) = (a-1)x^{-a} \cdot \mathbb{1}_{[1;+\infty)}$

1/ For what values of a, f_X is a probability density function.

2/ For these values of a, determine $\mathbb{E}(|X|)$.

Exercise 2

We throw a coin 3 times in a row.

1/ List all possible results by noting S for Stack and F for Face (example : SSF).

2/ Give the probability of the following events :
A "the draw has only stacks".
B "the draw has at least one Face".

Exercise 3

Let us toss a coin 3 times. Then $\Omega = \{0,1\}^3$, where 0 denotes "tail" and 1 denotes "head". For $k \in \{1,2,3\}$ let $H_k \in \mathcal{P}(\Omega)$ be the event that describes "head" in the $k-th$ toss, i.e. $H_k = \{\omega = (\omega_1, \omega_2, \omega_3) \in \Omega; \omega_k = 1\}$.

1/ explain the set $\bigcap H_k$, for $k \in \{1,2,3\}$

2/ Check if the family (H_1, H_2, H_3) is independent.

Exercise 4

Part 1

Let $(\Omega, \mathbb{P})$ a discrete probability space, and $(H_1, H_2, \cdots, H_n)$ a partition of Ω into n events.

1/ Show that, for $i = 1, 2, \cdots, n$ if A is a non zero probability event :

$$\mathbb{P}(H_i|A) = \frac{\mathbb{P}(A|H_i)\mathbb{P}(H_i)}{\sum_{j=1}^{n} \mathbb{P}(A|H_j)\mathbb{P}(H_j)} \tag{1}$$

2/ One disease M affects one in 1000 people in a given population. We have a blood test that detects M with 99% reliability when this disease is actually present. However, one gets also a false positive result for $0,2\%$ of healthy people tested.
What is the probability that a does anyone actually get sick when their test is positive ?
Hint : We note M all sick people, and T the set of people whose test is positive ; such that $\mathbb{P}(M) = 0.001$, $\mathbb{P}(T|M) = 0.99$, and $\mathbb{P}(T|M^c) = 0.002$.

Part 2

Let T be the decay time of a given radioactive particle. It is assumed that T follows a exponential distribution whit parameter $\lambda = log(4)$. What is the probability that T is between k and his successor $k+1$, where $k \in \{3, 6, 9, 12, 15, \cdots\}$?

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Knowledge Test 2 [2]

Exercise 1

Consider the following region $B = \{(s,t) \in \mathbb{R}^2, s \geq 0, t \geq 0; s^2 + t^2 \leq 1\}$ and let $Y = (Y_1, Y_2)$ be a random vector that is uniformly distributed on B.

a) Draw the region B.
b) Calculate the marginal density f_{Y_1}.
c) Determine the expectation of Y_1.
d) Determine $\mathbb{E}(Y_1|Y_2 = t)$ for $0 \leq t \leq 1$.

Exercise 2

Consider the sequence $(s^{|n|})_{n\in\mathbb{Z}}$ for $s \in \mathbb{R}$.

a/ For which parameters $s \in \mathbb{R}$ does there exist $c \in \mathbb{R}^\star$ such that the sequence $(cs^{|n|})_{n\in\mathbb{Z}}$ becomes a probability mass function? Also calculate c in these cases.

b/ Determine the probability of the event $A = \{2k, k \in \mathbb{N}^\star\} = \{2, 4, 6, 8, 10, \cdots\}$ for all suitable s.

Exercise 3

Let $\mathcal{M} \in \mathbb{R}^2$ be the right-angled trapezoid with vertices $(0;0); (2;0); (2;1); (1;1)$. We consider a continuous random vector $(X;Y)$ having an equal distribution on $\mathcal{M}$

1/ Write down the common probability density of $(X;Y)$.
2/ Determine the marginal probability density of Y .
3/ You can determine the expected value of Y.

Exercise 4

Consider a system consisting of 3 components. The system works if A and (B or C) work.

Let L be the random variable describing the lifetime of the system as a whole. The lifetime of the single components is described by the independent random variables L_k, $k \in \{A, B, C\}$. Each of the random variables L_k is exponentially distributed with parameter $\lambda > 0$.

a)- Denote by L_{BC} the lifetime of the subsystem consisting only of components B and C. Determine the cumulative distribution function of L_{BC}.

b)- The cumulative distribution function $F_L(t)$ of L is given as

$$\begin{cases} 0 \qquad for \qquad t < 0 \\ 1 - 2\exp(-2\lambda t) + \exp(-3\lambda t) \qquad for \qquad t \geq 0 \end{cases} \tag{2}$$

Calculate the expected lifetime of the system.

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